

SF2940 Probability Theory – Exercises

Lukas Schoug

Information

This is a list of exercises for the course SF2940. A subset of these will be solved during the exercise sessions and one or two exam questions might be chosen from here, so the student is recommended to try to solve all of the problems. Solutions will **not** be handed out per request. This list will be updated throughout the course, and the format is not complete at this time.

When referring to the course literature in this document, we write [K] for *Lecture Notes: Probability and Random Processes at KTH* by Timo Koski and [G] for *An Intermediate Course in Probability* by Allan Gut.

Week 1

Exercise session 1

- Prove that if $\mathcal{F}$ is a σ -algebra, then the following holds. If $A_1, A_2, \dots \in \mathcal{F}$, then $\bigcap_{j=1}^{\infty} A_j \in \mathcal{F}$. (Hint: use De Morgan's laws.)
- Exercise 1.12.2.6 in [K].
- Exercise 1.12.2.7 in [K].
- Exercise 1.12.2.9 in [K].
- Exercise 1.12.3.6 in [K].
- Exercise 1.12.3.10 in [K].

Exercise session 2

Prioritised during session

- Exercise 2.6.2.4 in [K].
- Exercise 2.6.3.6 in [K].
- Exercise 2.6.3.15 in [K].

- Exercise 2.6.3.17 in [K]. **ERROR:** In some versions the answer to (b) is wrong. The answer to (b) should be

$$f_{X-Y}(v) = \frac{1}{2(1+|v|)^2}, \quad -\infty < v < \infty.$$

- Exercise 2.6.3.20 in [K].

Recommended

- Exercise 2.6.2.1 in [K].
- Exercise 2.6.2.2 in [K].
- Exercise 2.6.2.5 in [K].
- Exercise 2.6.3.1 in [K].
- Exercise 2.6.3.4 in [K].
- Exercise 2.6.3.5 in [K].
- Exercise 2.6.3.10 in [K].
- Exercise 2.6.3.12(a,b) in [K]. **ERROR:** Note that in (b), X/Y and Y should be independent, not X and X/Y .
- Exercise 2.6.3.21 in [K].
- Problem 1.24 in [G]. **Comment:** Compare with Exercise 2.6.3.12 in [K].

Week 2

Exercise session 3

Prioritised during session

- Exercise 2.6.5.2 in [K].
- Exercise 2.6.5.5 in [K].
- Exercise 2.6.5.6 in [K].
- Exercise 3.8.2.3 in [K].
- Exercise 3.8.2.5 in [K].
- Exercise 3.8.2.13 in [K].

Recommended

- Exercise 3.8.2.10 in [K].
- Exercise 3.8.2.14 in [K].
- Exercise 3.8.2.15 in [K].
- Problem 2.9 in [G].
- Problem 2.12 in [G].
- Problem 2.14 in [G].
- Problem 2.19 in [G].

Exercise session 4

Prioritised during session

- Exercise 3.8.3.5 in [K].
- Exercise 3.8.3.6 in [K].
- Exercise 3.8.3.8 in [K].
- Exercise 3.8.3.10 in [K].
- Exercise 3.8.3.14 in [K].
- Exercise 3.8.4.3 in [K].

Recommended

- Exercise 3.8.3.1 in [K].
- Exercise 3.8.3.2 in [K].
- Exercise 3.8.3.9 in [K].
- Exercise 3.8.3.11 in [K].
- Exercise 3.8.4.8 in [K].
- Exercise 3.8.4.13 in [K].

Week 3

Exercise session 5

Prioritised during session

- Problem 2.6.25 in [G].
- Problem 2.6.33 in [G].
- Problem 3.8.2 in [G].
- Exercise 5.8.1.1 in [K].
- Exercise 5.8.1.4 in [K].
- Exercise 5.8.1.5 in [K].
- Exercise 5.8.2.6 in [K].

Recommended

- Problem 2.6.22 in [G].
- Problem 2.6.23 in [G].
- Exercise 5.8.1.2 in [K]. (Focus on parts (a) and (b). Exercise 5.8.1.1 gives the necessary information about the Pascal distribution.)
- Problem 3.8.1 in [G].
- Problem 3.8.3 in [G].
- Problem 3.8.6 in [G].
- Problem 3.8.8 in [G].

Exercise session 6

Prioritised during session

- Exercise 5.8.2.7 in [K].
- Exercise 4.7.1.2 in [K].
- Exercise 4.7.1.3 in [K].
- Exercise 4.7.1.6(a) in [K].
- Exercise 4.7.1.7 in [K].
- Exercise 4.7.2.1 in [K].
- Exercise 5.8.3.12 in [K].

Recommended

- Exercise 4.7.1.1 in [K].
- Exercise 4.7.1.5 in [K]. Hint: use the result of Exercise 4.7.1.4 in [K].
- Exercise 4.7.1.6(b) in [K].
- Exercise 4.7.1.8(a) in [K].
- Problem 3.8.19 in [G].
- Problem 3.8.21 in [G].

Week 4

Exercise session 7

Prioritised during session

- Problem 3.8.18 in [G].
- Problem 3.8.22 in [G].
- Problem 6.8.1 in [G]. (**NOTE:** Assume that $X_1, X_2, \dots$ are independent.)
- Problem 6.8.6 in [G]. (**ERROR:** The statement should have $Y_n = \min\{X_1, X_2, \dots, X_n\}$.)
- Exercise 6.8.1.2 in [K].

Recommended

- Problem 6.8.2 in [G].
- Problem 6.8.8 in [G].
- Problem 3.8.14 in [G].
- Problem 3.8.23 in [G].

Week 5

Exercise session 8

Prioritised during class

- Exercise 6.8.1.3 in [K].
- Exercise 6.8.1.15 in [K].
- Exercise 6.8.1.16 in [K].

- Exercise 6.8.1.17 in [K].
- Exercise 6.8.2.1(a) in [K].
- Exercise 6.8.2.7 in [K].
- Exercise 6.8.3.1 in [K].

Recommended

- Exercise 6.8.1.2 in [K].
- Exercise 6.8.1.6 in [K].
- Exercise 6.8.1.7 in [K].
- Exercise 6.8.1.8 in [K].
- Exercise 6.8.1.12 in [K].
- Problem 6.8.13 in [G].

Exercise session 9

Prioritised during class

- Problem 6.8.5 in [G].
- Problem 6.8.15 in [G].
- Problem 6.8.28 in [G].
- Problem 6.8.31 in [G].
- Problem 6.8.23 in [G].
- Problem 6.8.32 in [G].

Recommended

- Problem 6.8.11 in [G].
- Problem 6.8.18 in [G].
- Problem 6.8.20 in [G].
- Problem 6.8.21 in [G].
- Problem 6.8.34 in [G].
- Problem 6.8.43 in [G].

Week 6

Exercise session 10

Prioritised during class

- Exercise 8.5.1.3 in [K].
- Exercise 8.5.1.10 in [K].
- Exercise 8.5.1.13 in [K].
- Exercise 8.5.1.15 in [K].
- Problem 5.10.10 in [G].
- Problem 5.10.23 in [G].

Recommended

- Exercise 8.5.1.11 in [K].
- Exercise 8.5.1.16 in [K].
- Exercise 8.5.1.17 in [K].
- Problem 5.10.9 in [G].
- Problem 5.10.18 in [G].

Week 7

Exercise session 11

Solve the re-exam 2024 and perhaps parts of the re-exam 2023.